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The Influence of Primary Air Jets on Flow and Pollutant Emissions Characteristics within a Model Gas Turbine Combustor

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The present work studies the effect of utilizing counter swirling flow on pollutant emission characteristics at different equivalence ratios and inlet air temperatures. A PIV system is employed to study the aerodynamic interactions between the recirculation zone of a model gas turbine combustor and the primary air jets. Centerline measurements of emissions generated during combustion of Jet-A liquid fuel in a swirl cup assembly provide more insight into nitric oxide and carbon monoxide formation/destruction processes. Results showed that misplacement of the primary air jets may cause production of higher amounts of carbon monoxide and nitric oxides. Besides, over penetrating primary air participates in thermal quenching for carbon monoxide.

Nomenclature

A	=	amplitude of oscillation
a	=	cylinder diameter
C_p	=	pressure coefficient
Cx	=	force coefficient in the x direction
Cy	=	force coefficient in the y direction
c	=	chord
dt	=	time step
Fx	=	X component of the resultant pressure force acting on the vehicle
Fy	=	Y component of the resultant pressure force acting on the vehicle
f, g	=	generic functions
h	=	height

I. Introduction

DILUTION jets play a vital role to control the combustion temperature, enhance mixing, and achieve satisfactory combustion performance within a gas turbine combustor. In the primary zone, dilution jets are required to provide more uniform mixing, promoting efficient combustion and minimizing pollutant formation. The attainment of a satisfactory temperature distribution in the exhaust gases is mainly dependent on the degree of mixing between air and combustion products in the dilution zone.

Extensive work have been carried out to study and characterize the dilution jet mixing processes in the dilution zone of the combustor [1-5] but very few studies were concerned with the primary zone which is believed to be the

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main contributor for pollutant emissions in all combustor zones. Unfortunately, there is no clear strategy for the placement of the transverse air jets in the primary zone. Dodds and Bahr [6] emphasized that interactions between swirler, primary jets, and cooling flows are known to be important, but the effects of these interactions on emissions and performance have not been systematically studied. Lefebvre [7] stated that as a general rule, maximum stability is achieved by injecting primary air through a small number of large holes. Although no firm guidelines have been laid down for the optimal number of liner holes for annular combustors, one opposing pair of holes per fuel injector should be regarded as the absolute minimum; twice that number would be preferable.

Gogineni et al. [8], pointed out that many aircraft engine combustors use transverse air jets introduced through the liners to provide a nominal termination of the primary zone. Richards and Samuelsen [9, 10] showed that the relative positions of the transverse jets to the swirl injection plane are of paramount importance. They suggested placement of the transverse jets in the region of half to one duct diameter for optimum injection. Mellor et al [11] suggested that the placement of the first row of penetration air jets will largely determine the emissions within the combustor. Noyce et al [12] showed that the primary air addition methods have a dominant influence on pollutant levels at the combustor exit. On the contrary, in a study by Gogineni et al [8], it was concluded that most of the NO_x was generated from stoichiometric burning in interfaces around the combustion air jet system. Minimum total emissions were produced when all of the reactants are introduced through the combustor dome. However air jets were necessary for improving lean blowout stability and shorter flame length. Sturgess et al [13] concluded that minimum overall pollution is generated when air jets are positioned 0.5 dome heights downstream from the dome.

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II. Experimental Setup

A. Test Facility

A schematic diagram of the test facility is shown in Fig. 2. The air system consists of a compressor, a dryer, a 0.5 micron particle filter, a pressure regulator, and an air storage tank. An ASME orifice flow meter is utilized to measure airflow rate into the system. A 39 kW heater can be used to preheat the air per test requirements. The air passes through a honeycomb flow conditioner to the test section. A separate air line equipped with a flowmeter, regulator, and pressure transducers is utilized to control the amount of air flow through the transverse jets. The fuel system includes a fuel tank, a fuel pump, a fuel filter, a coriolis flowmeter, and a needle valve for controlling the fuel flow rate. Fuel is injected by a hollow cone simplex atomizer with 90-deg spray angle mounted in the center of the swirler. The swirler used in this study is designed to generate two coaxial, counter swirling airstreams. The swirler flare exit diameter D is 40mm. The detailed design of this swirl-cup has been fully described elsewhere (thesis).

The combustion chamber used in this study is made of a combination quartz glass, for optical access, and stainless steel plates. The cross section of the chamber is a square with a side dimension of 76.2mm. The chamber height is about 330mm. In this study, the chamber is designed such that it can:

- Allow for the study of the effect of transverse air jets on aerodynamics and emissions characteristics,
- Allow a gas sampling probe to traverse inside the chamber in order to study the effect of different parameters on emissions within the primary zone.

A combustion chamber with the swirl-cup assembly is shown in Fig.2. In this case the chamber is designed such that primary air jets location as well as jet diameter can be varied. Eight jet heights can be examined; four of them are shown in the figure and the other four locations can be obtained by turning the primary jets holder 180°. Thus, jet heights can be varied in the range of 25mm to 114mm with a step of 12.7mm. Moreover, the air jet diameter can be varied by using jet connectors with different inner diameters. However, in this study only transverse air jets with a diameter of 12.7mm were utilized. When no dilution air is allowed to be injected to the chamber, the primary jets holder is plugged.

In order to study the transient as well as average flow aerodynamics within the primary zone of a gas turbine combustor, a Particle Image Velocimetry (PIV) system is utilized. The current system can provide two- or three-dimensional velocity maps in flows using whole field techniques based on imaging the light scattered by small particles in the flow illuminated by a laser light sheet. For non-reacting flow measurements, the flow was seeded, using a six jet TSI flow seeder, with propylene Glycol which provides particles in the size range of 1-3 μm.

Double Pulsed Laser ND: YAG laser (NewWave, Solo PIV) with pulse energy of up to 120 mJ per pulse at 15 HZ and a wave length of 532 nm was utilized to illuminate the seeding particles with short time difference.

A double frame-double exposure CCD Camera (Lavision - Image Intense) is placed perpendicular to the illuminated swirl-cup center plane in the case of 2D-PIV measurements as depicted in Fig. 2. The Camera delivers 12 bit digital images with a resolution of 1376 x 1040 pixels per frame. It features a built in electronic shutter with

exposure times as short as 500ns and can provide 5 double frames per second. A Nikon D-type lens with 50 mm focal length and maximum aperture of f/1.8 was placed in front of the CCD camera. A bandpass filter with 532 ±3nm is placed in front of the camera lens to suppress the background light in the second frame.

One CCD camera was setup to capture a field of view of 75mm width x 100 mm height. However, a larger field of view was required to cover the whole recirculation zone region and part of the post recirculation zone. To achieve this large field of view without altering the resolution, it was decided to use two CCD cameras side-by-side as presented in Fig. 2. In this case a special calibration for both cameras in which the calibration of camera 2 is mapped to the calibration of camera 1. The resulting vector fields from both cameras are then merged together to form one enlarged vector field.

The scattered light from first and second exposure of the particles within this field of view is recorded in two different images which are then divided to small sub-areas called interrogation windows. Using statistical correlation techniques (cross correlation) one local displacement vector is determined for each interrogation window. The window size selected for image processing is 32 x 32 pixels with a 50% overlap in each direction. As a result, this approach led to a measurement grid with a spatial resolution of about 1.5 mm x 1.5 mm.

Both the main flow through the swirl-cup and the flow through the primary jets are seeded using propylene glycol. Frame straddled pairs are acquired with laser pulse delay (dt) in the range of 5-8 μs. The main flow structure is evaluated by ensample-averaging 200 successive double frames.

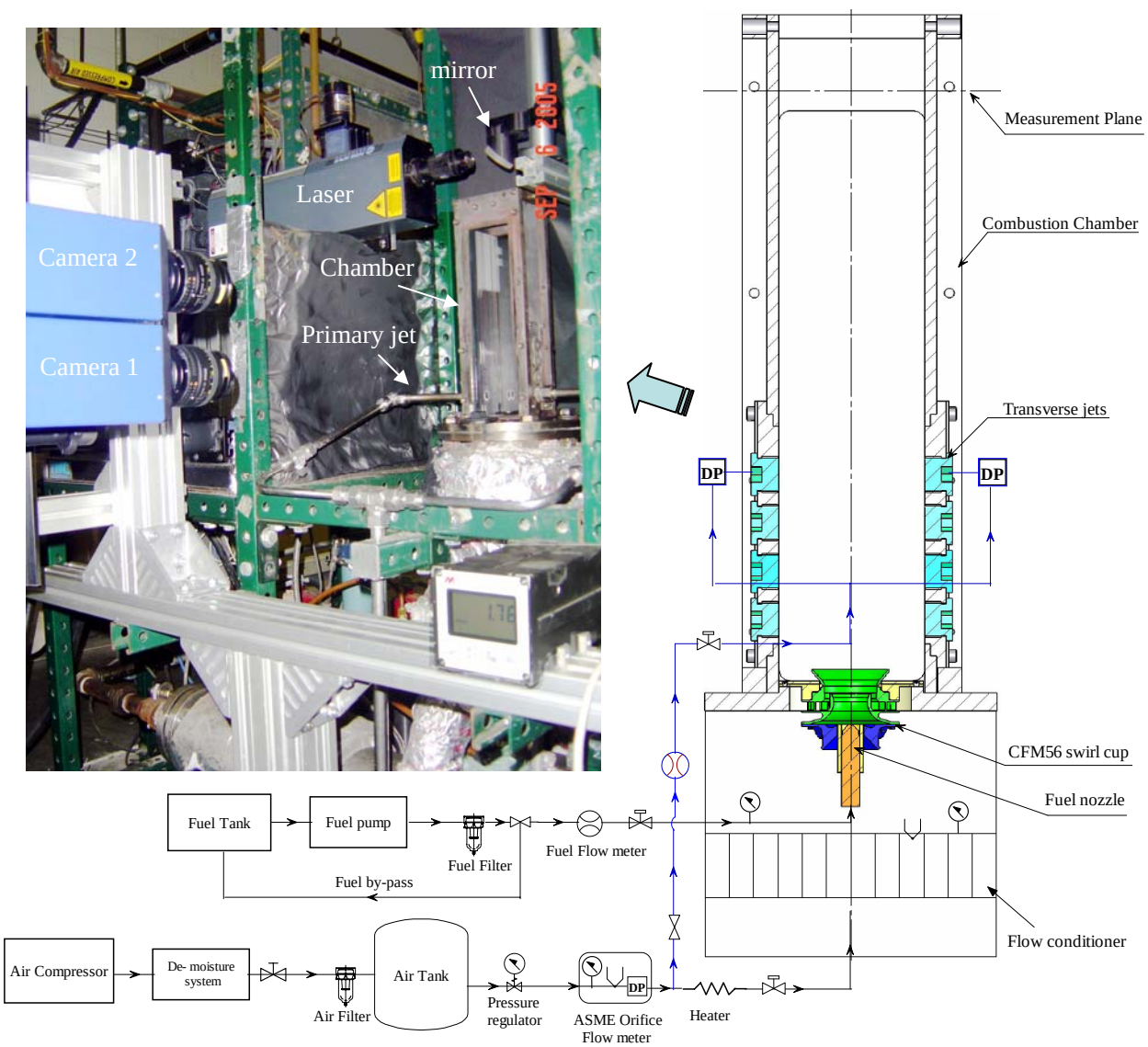

Figure 2. PIV experiment setup to study primary air jets aerodynamics

III. Results

Since coupling between fluid dynamics and chemical processes within the combustion chamber provide a major factor governing the production of pollutant emissions, it was decided to check the aerodynamic influence of the primary jets on the main swirling flow. Non-reacting flow measurements were considered because the main objective is to investigate the trends of the flow-field at different primary jets conditions. The experimental setup for this case was presented in Fig. 2.14.

Two main parameters are studied these are the jet location and mass ratio. Since there is no combustion, the mass flow through the swirler is kept fixed to a value corresponding to a pressure drop of 4.0% across the swirler. The mass ratio is varied by controlling the mass flow rate through the primary jets. The test parameters are provided in table 5.5.

PIV results for the base line case, where no air is allowed to flow through the primary jets, is presented in Fig. 5.13. The swirling air departing from the swirler remains attached to the swirl-cup conical sleeve expands away from the centerline and impinges on the chamber walls at about 25 mm downstream of the swirler exit plane. Upon impingement, the main flow velocity weakens; the outward radial velocity decays rapidly and the axial velocity drops by about 40% if compared to that just departing the swirler. Due to confinement, the recirculation zone is extended to about Z =104 mm and it occupies about 45% of the chamber size at about Z =35 mm

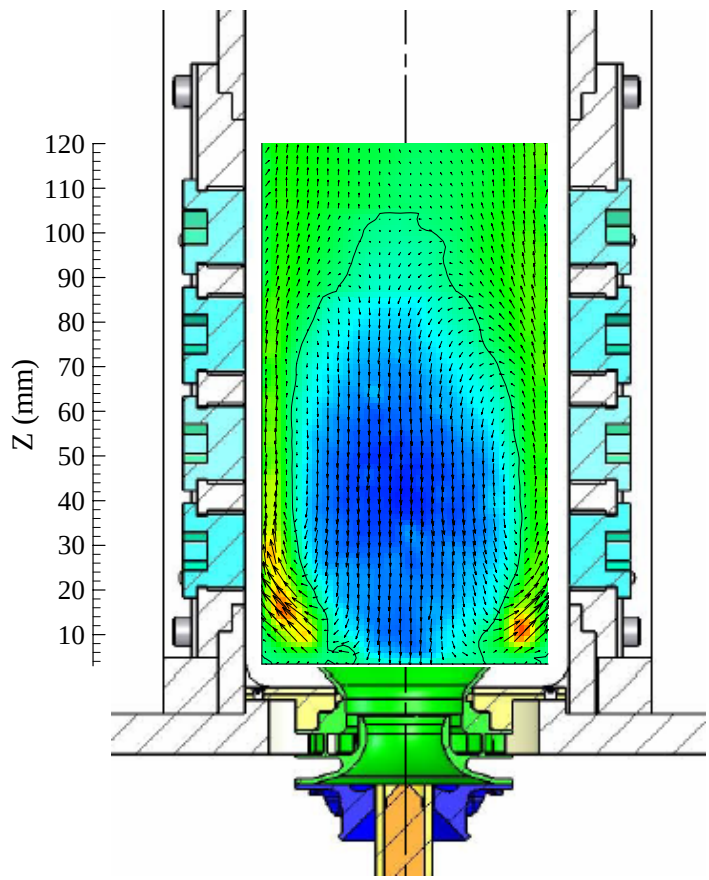

Figure III.1 Axial velocity contours and velocity vectors for the baseline case (MR=0.0)

The influence of primary air jets is presented in figures 5.14-5.16 for different jet heights. Within each jet location, the mass ratio was varied from 0.07 to 0.4 to investigate the effect of the jet strength and how it interacts with the recirculation zone.

Figure 5.14 describes the case of primary air injection 25 mm downstream of the swirler exit. Significant changes in velocity profiles are observed when varying the mass ratio. In this case one can think of the recirculation zone as a balloon, holding it by hand and pressuring on it gently, the balloon contracts from its side but elongates from the other sides. As the pressure increase, the balloon elongates more but if the pressure is more than it withstands, it ruptures.

For the lowest mass ratio considered (MR= 0.07), the primary air deflects right away by the strong main stream issuing from the swirl-cup. The jet deflection angle is about 50° measured from the jet centerline indicating that the main swirling flow overpowers the weak jets. The recirculation zone is affected by the presence of the jets. It contracts from its side by the incoming air jets. For instance, its width is shortened from about 55 mm to 42 mm at Z = 45 mm. However, the recirculation zone height is extended from 104 mm to 112 mm. When the primary jets mass flow rate is increased to MR = 0.1, it become more resistant to the main flow. The jet deflection angle reduces to 40° but the recirculation zone contracts more on its side. The width of the recirculation zone reduces to 31 mm but its height extends to 118 mm. Increasing the mass ratio to 0.15 leads to even more contraction to the recirculation zone side. The jets become

more resistant to the main flow, the jet deflection angle reduces to 30° and a corner vortex forms just downstream of the jet.

If the mass ratio is increased beyond 0.15, the recirculation zone detaches and its overall size shrinks with increasing the mass ratio. As shown in Fig. 5.14 d-f, the jets are more resistant and overpower the main flow. This behavior is noticeable when observing the jet deflection angle which decreases significantly with increasing MR. The two opposing jets penetrate through the recirculation zone and impinge on each other rupturing the recirculation zone. A comparison between the cases considered is provided in table 5.6. These observations can explain why flame stability deteriorates when injecting primary air with mass ratios greater than 0.16.

Mass Ratio	0	0.07	0.1	0.15	0.2	0.3	0.4
Deflection angle (deg.)	-	50	40	30	18	9	6
R.Z. width @ Z=45mm	55	42	31	20	-	-	-
R.Z. height (mm)	104	110	118	119	41.5	35	33

Table III.1 Influence of jet mass ratio on recirculation zone properties for primary air placed 25mm above the swirler exit plane

These results also clarify why there is a trade off between CO emissions and primary jets mass ratio. Tracking the primary jets path at each mass ratio studied, it can be observed that for low mass ratios (or high deflection angles) the jet air takes longer time prior to penetrating the recirculation zone thus mixing is enhanced between the jets and the main swirling flow. This primary air becomes part of the combustion process and its amount is not sufficient to form a spot with cold temperature that can deteriorate the oxidation process of CO to CO2. On the other hand, for high jet mass ratios (MR>0.2), the primary air overpowers and does not take time to mix with the main flow. This primary air penetrates through the recirculation zone forming a region of cold temperature that quenches the reaction and suppresses the oxidation of CO to CO2.

Based on these results one can suggest that the optimum mass ratio in this case is 0.15 because the recirculation zone is still preserved, and the two opposing jets are not impinging on each other forming a cold spot that might quench the chemical reaction.

Referring to the baseline case where no jet air was admitted, Fig. 5.13, the main swirling air velocity decay very rapidly upon impingement on the chamber walls which happens at about Z=25 mm. It is expected that if the primary air jets are located far from the impingement location it will face less resistance by the main flow to penetrate through the recirculation zone. This anticipation is demonstrated in figures 5.15 and 5.16 where the primary air was injected at Z =50 mm and Z =76mm respectively.

The jet deflected angle can be used as a measure of the resistance of the primary jet to the main flow. As shown in table 5.7, the deflection angle drops significantly as the jet is placed farther from the plane where the main flow impinges on the chamber. As the deflection angle decrease indicating a strong jet, mixing with the main flow deteriorates. In combustion cases this is not a favorable situation because the jets might act only as a heat sink that can quench the reaction leading to higher CO and UHC at the combustor exhaust.

Mass Ratio	Deflection angle at different jet heights		
	Z =25 mm	Z =50 mm	Z =76 mm
0.07	50	22	35
0.1	40	18	10
0.15	30	8	3
0.2	18	3	1
0.3	9	0	0
0.4	6	0	0

Table III.2 Deflection angle at different primary air injection locations

The recirculation zone shortens with increasing the mass ratio but since the primary air is injected farther downstream, the recirculation zone size is still sufficient to sustain flame stability. This explains why primary air jets injected at Z =50, 76 and 101 mm are superior to that injected at 25 mm in terms of flame stability.

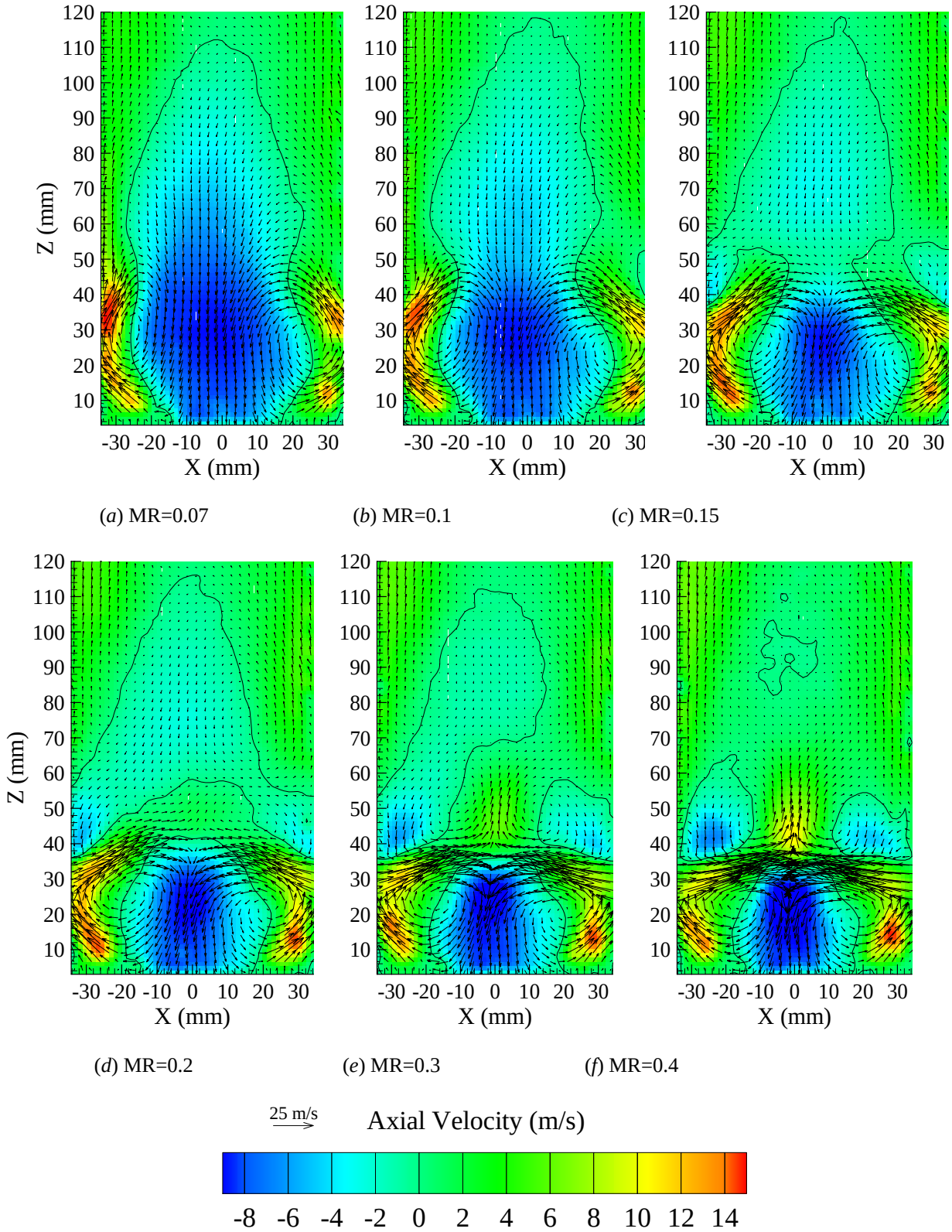

Figure III.2 Axial velocity contours and velocity vector plots for primary air injection located 25 mm above swirler exit at different mass ratios

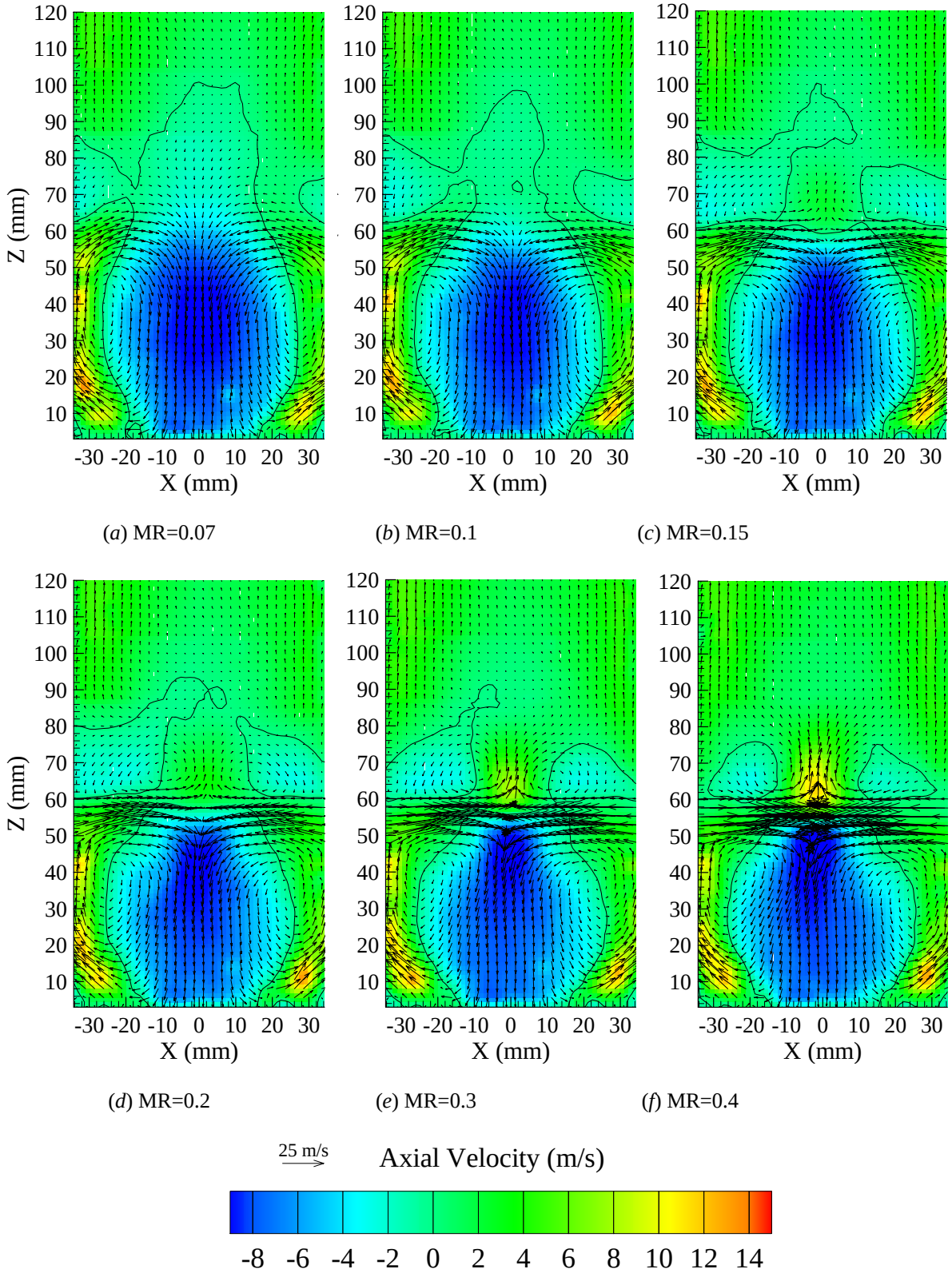

Figure III.3 Axial velocity contours and velocity vector plots for primary air injection located 50 mm above swirler exit at different mass ratios

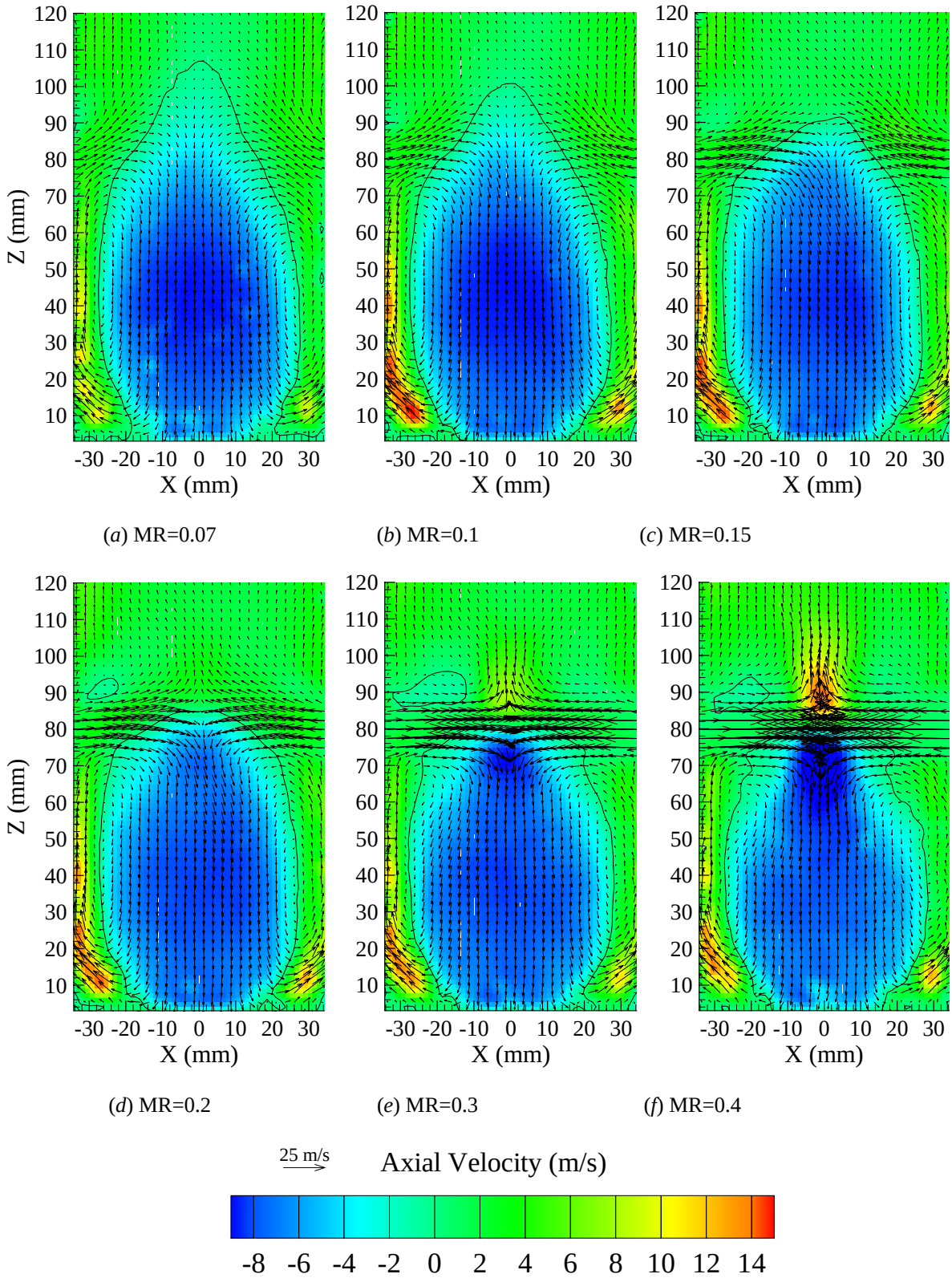

Figure III.4 Axial velocity contours and velocity vector plots for primary air injection located 76 mm above swirler exit at different mass ratios

IV. Conclusion

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